Paper id: 250619

Subject Code: BAU061

Roll No: | | | | | | | | | | | | | |

BTECH

(SEM VI) THEORY EXAMINATION 2024-25

AUTOMOTIVE ELECTRICAL & ELECTRONICS

TIME: 3 HRS **M.MARKS: 70**

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt *all* questions in brief. **02 x 7 = 14**

Q no.	Question	CO	Level
a.	Define Ohm's Law and Kirchhoff's Law.	1	K1
b.	List any four types of automotive batteries.	1	K1
c.	State the main function of a starter motor.	2	K1
d.	Differentiate between D.C. and A.C. generator.	2	K1
e.	What is a spark plug and its role in ignition?	3	K1
f.	Write any two advantages of CAN bus system.	4	K1
g.	List any two types of automotive sensors.	5	K1

SECTION B

2. Attempt any *three* of the following: **07 x 3 = 21**

a.	Explain the construction and working of a Lead Acid Battery.	1	K1,K2
b.	Describe the construction and types of starter motors used in vehicles.	2	K1,K2
c.	Discuss the role and working of alternators in charging systems.	2	K1,K2
d.	Explain the working principle of distributor-less ignition system	3	K1,K2
e.	Describe the architecture and functions of automotive electronic systems.	4	K1,K2

SECTION C

3. Attempt any *one* part of the following: **07 x 1 = 07**

a.	Explainbattery charging techniques and tests conducted on automotive batteries.	1	K1,K2
b.	Discuss the maintenance of different types of automotive batteries.	1	K1,K2

4. Attempt any *one* part of the following: **07 x 1 = 07**

a.	Explain with diagram the starter drive mechanism and its components	2	K1,K2
b.	Describe voltage and current regulation in automotive charging systems.	2	K1,K2

5. Attempt any *one* part of the following: **07 x 1 = 07**

a.	Explain various ignition triggering devices used in modern vehicles.	3	K1,K2
b.	Describe the centrifugal and vacuum advance mechanisms in ignition systems.	3	K1,K2

6. Attempt any *one* part of the following: **07 x 1 = 07**

a.	Explain various bus systems used in automotive electronics with suitable examples.	4	K1,K2
b.	Describe the working and applications of ABS and ECM control units	4	K1,K2

7. Attempt any *one* part of the following: **07 x 1 = 07**

a.	Explain in brief about different types of sensors used in driver assistance systems.	5	K1,K2
b.	Describe working principles of LIDAR and Radar sensors in automotive systems.	5	K1,K2

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